SUMMARY

Inverse Trigonometric Functions

Inverse sine function	Inverse cosine function	Inverse tangent function	Inverse cosecant function	Inverse secant function	Inverse cot function
Domain $[-1,1]$	Domain $[-1,1]$	Domain $\mathbb{R}$	Domain $(-\infty,-1]\cup[1,\infty)$	Domain $(-\infty,-1]\cup[1,\infty)$	Domain $\mathbb{R}$
Range $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$	Range $[0,\pi]$	Range $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$	Range $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]-\{0\}$	Range $[0,\pi]-\left\{\frac{\pi}{2}\right\}$	Range $(0,\pi)$
not a periodic function	not a periodic function	not a periodic function	not a periodic function	not a periodic function	not a periodic function
odd function	neither even nor odd function	odd function	odd function	neither even nor odd function	neither even nor odd function
strictly increasing function	strictly decreasing function	strictly increasing function	strictly decreasing function with respect to its domain.	strictly decreasing function with respect to its domain.	strictly decreasing function
one to one function	one to one function	one to one function	one to one function	one to one function	one to one function

Properties of Inverse Trigonometric Functions

Property-I

(i) $\sin^{-1}(\sin\theta)=\theta$, if $\theta\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

(ii) $\cos^{-1}(\cos\theta)=\theta$, if $\theta\in[0,\pi]$

(iii) $\tan^{-1}(\tan\theta)=\theta$, if $\theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

(iv) $\operatorname{cosec}^{-1}(\operatorname{cosec}\theta)=\theta$, if $\theta\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\setminus\{0\}$

(v) $\sec^{-1}(\sec\theta)=\theta$, if $\theta\in[0,\pi]\setminus\left\{\frac{\pi}{2}\right\}$

(vi) $\cot^{-1}(\cot\theta)=\theta$, if $\theta\in(0,\pi)$

Property-II

(i) $\sin\left(\sin^{-1}x\right)=x$, if $x\in[-1,1]$

(ii) $\cos\left(\cos^{-1}x\right)=x$, if $x\in[-1,1]$

(iii) $\tan\left(\tan^{-1}x\right)=x$, if $x\in\mathbb{R}$

(iv) $\operatorname{cosec}\left(\operatorname{cosec}^{-1}x\right)=x$, if $x\in\mathbb{R}\setminus(-1,1)$

(v) $\sec\left(\sec^{-1}x\right)=x$, if $x\in\mathbb{R}\setminus(-1,1)$

(vi) $\cot\left(\cot^{-1}x\right)=x$, if $x\in\mathbb{R}$

Property-III (Reciprocal inverse identities)

(i) $\sin^{-1}\left(\frac{1}{x}\right)=\operatorname{cosec}x$, if $x\in\mathbb{R}\setminus(-1,1)$.

(ii) $\cos^{-1}\left(\frac{1}{x}\right)=\sec x$, if $x\in\mathbb{R}\setminus(-1,1)$

(iii) $\tan^{-1}\left(\frac{1}{x}\right)=\begin{cases}\cot^{-1}x & \text{if } x>0\\ -\pi+\cot^{-1}x & \text{if } x<0.\end{cases}$

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